

Experiment - 8

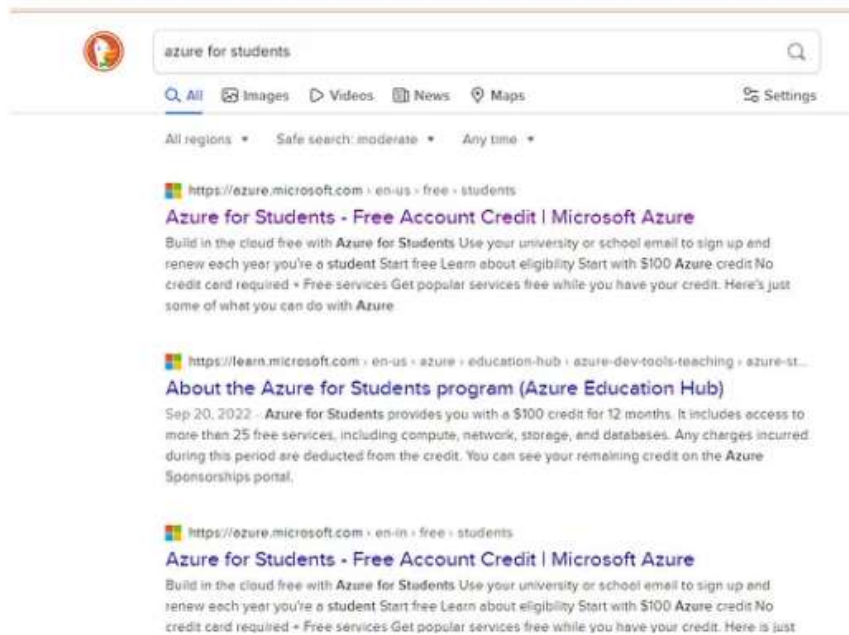
1) Aim/Objective:

Case Study on Microsoft Azure

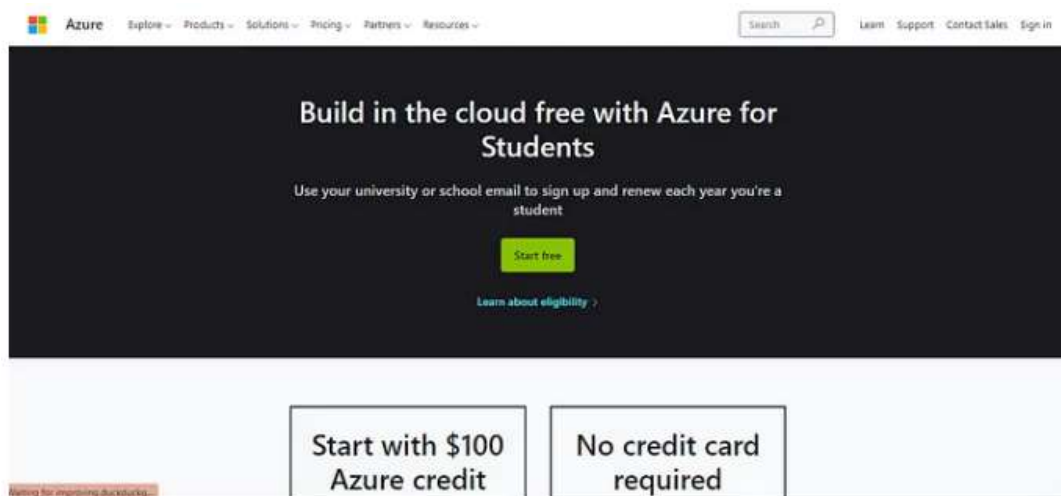
2) Procedure:

Part 1: Creating an Azure Account

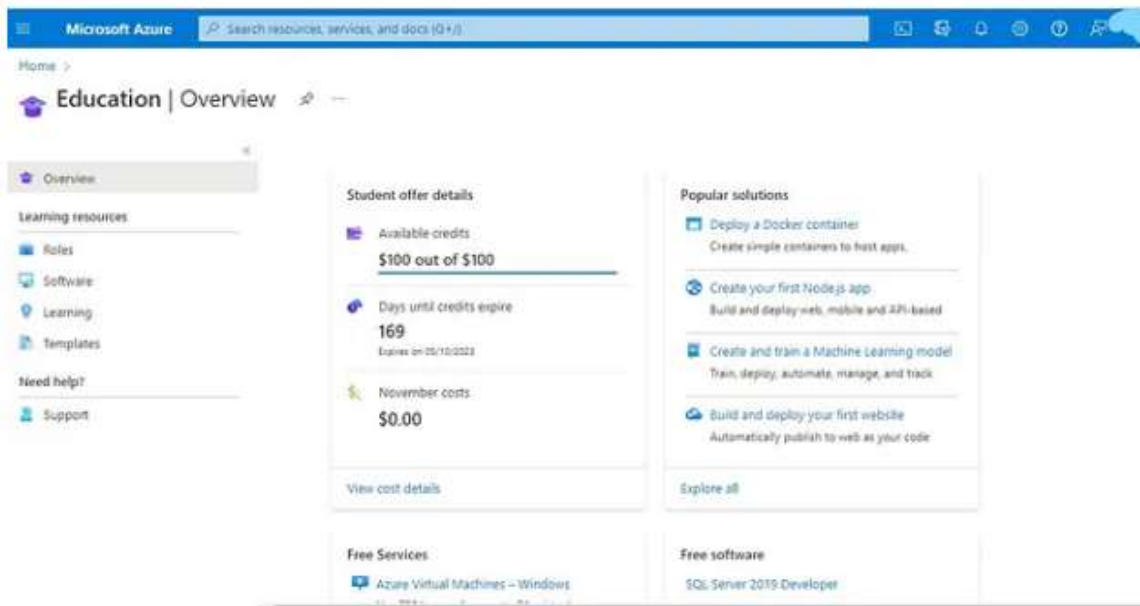
1. **Access the Azure Portal:** In a web browser, go to Azure for students, click on the first link you see <https://azure.microsoft.com/en-in/free/students>



2. **Start Free:** You are going to select START FREE and then sign in with your college email. For those just signing in for the first time, you'll have to verify your university as a code will be sent to your college email.



3. **Microsoft Account:** To access Azure for Students, you'll need a Microsoft account. Sign in with your existing account or create one if you don't have it yet. After filling in all the necessary details, your Azure portal is ready and you can start exploring all the benefits.
4. **Verify Your Student Status :**



Part 2: Exploring the Azure Portal

1. **Navigation:** Familiarize yourself with the Azure portal's interface. Pay attention to:
 - **Left-hand menu:** Provides access to various Azure services and resources.
 - **Search bar:** Allows you to quickly find specific services, resources, or documentation.
 - **Dashboard:** A customizable view of your frequently used resources and important information.
 - **Top bar:** Contains global settings, notifications, and your account information.
2. **Creating a Resource Group:**
 - Click on "Resource groups" from the left-hand menu or search for it in the search bar.
 - Click on "Create" and provide a name and region for your resource group.
 - Click "Review + create" and then "Create". Resource groups are logical containers for your Azure resources.
3. **Creating a Virtual Machine:**
 - Click on "Virtual machines" from the left-hand menu or search for it.
 - Click on "Create" and select "Azure virtual machine".
 - Configure the virtual machine settings:
 - **Resource group:** Select the resource group you created.
 - **Virtual machine name:** Give your VM a name.
 - **Region:** Choose a region close to you.
 - **Image:** Select an operating system (e.g., Ubuntu Server, Windows Server).
 - **Size:** Choose a VM size (you can start with a small, free tier eligible size).
 - **Administrator credentials:** Set a username and password or choose SSH key authentication.
 - **Inbound rules:** Configure network access (e.g., allow SSH or RDP).
 - Click "Review + create" and then "Create". Wait for the VM to be deployed.
4. **Creating a Storage Account:**
 - Click on "Storage accounts" from the left-hand menu or search for it.
 - Click on "Create".
 - Configure the storage account settings:
 - **Resource group:** Select your resource group.
 - **Storage account name:** Give your storage account a unique name.
 - **Region:** Choose the same region as your VM.
 - **Performance:** Select "Standard".

- **Redundancy:** Choose "Locally-redundant storage (LRS)".
 - Click "Review + create" and then "Create". Wait for the storage account to be deployed.
5. **Exploring Azure Active Directory (Azure AD):**
- Search for "Azure Active Directory" in the search bar.
 - Explore the features of Azure AD, such as users, groups, and applications. Azure AD is Microsoft's cloud-based identity and access management service.

Part 3: Managing Resources

1. **Start and Stop Virtual Machine:**
 - Navigate to your virtual machine in the Azure portal.
 - Use the "Start" and "Stop" buttons to manage the VM's power state.
2. **Access Storage Account:**
 - Navigate to your storage account.
 - Explore the different storage services like Blob storage, File storage, Queue storage, and Table storage.
3. **Monitor Resources:**
 - Use the "Overview" section of your resources to monitor their status, metrics, and logs.\

References:

- <https://medium.com/@oluseyejeremiah/creating-a-student-azure-account-ad1ee7797922>
- <https://gopalkumr.medium.com/microsoft-azure-a-step-by-step-guide-to-getting-student-credits-030204aa949d>